Project Design Phase-I
Solution Architecture

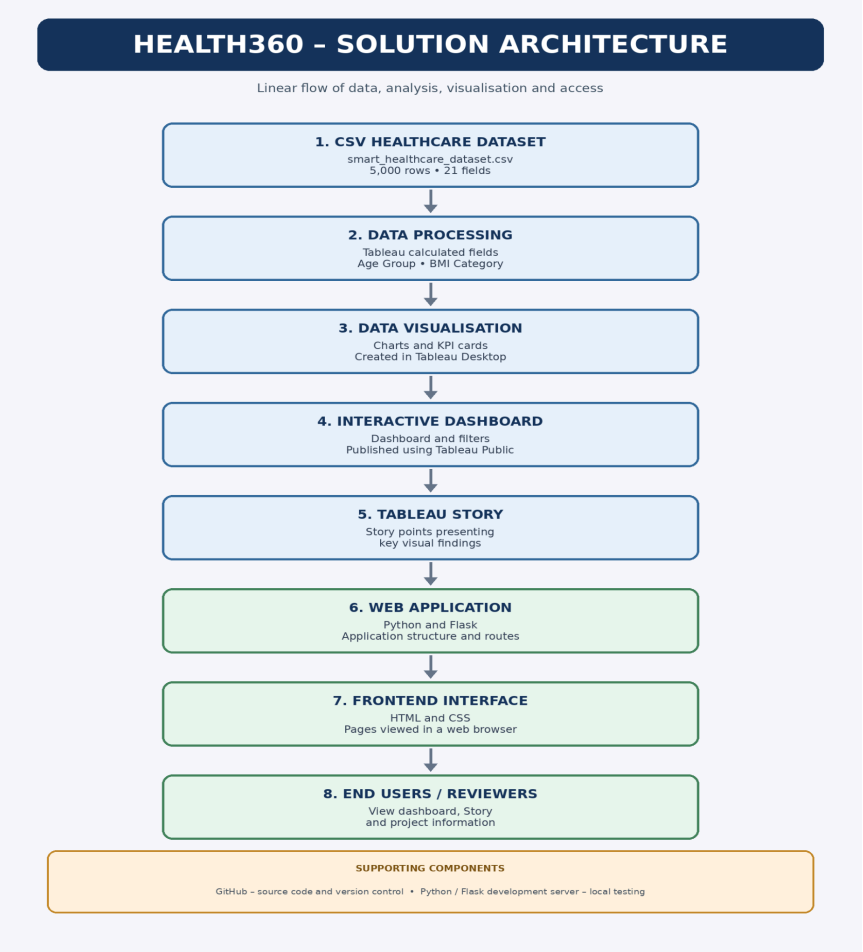
Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Healthcare Dataset	Stores patient information such as Age, Gender, BMI, Blood Pressure, Exercise Level, Alcohol, Smoking, Diabetes, Heart Disease, and Stroke.	CSV
Data Processing	Handles data preparation, calculated fields, categorisation, and aggregation required for healthcare analysis.	Tableau Calculated Fields
Data Visualization	Creates charts, graphs, KPI cards, and heatmaps to represent healthcare information visually.	Tableau Desktop
Interactive Dashboard	Displays patient statistics and healthcare visualisations with filters for Age Group, Gender, and BMI Category.	Tableau Desktop / Tableau Public
Tableau Story	Organises visualisations into a sequence to present healthcare observations and findings.	Tableau Desktop / Tableau Public
Web Application	Provides a website with Home, About, Dashboard, Story, and Contact sections to introduce the Health360 project.	Python, Flask
Frontend Interface	Designs the website layout, navigation menu, headings, buttons, and visual elements.	HTML, CSS
Version Control	Stores and manages project source code, dataset files, and documentation.	GitHub
Development Environment	Provides the environment for writing, running, and testing the Flask application.	Python, Flask Development Server, Web Browser